**Payload: REASON (Radar for Europa Assessment and Sounding: Ocean to Near-surface)**

REASON is a dual-frequency ice-penetrating radar sounder currently flying aboard NASA's Europa Clipper spacecraft. The instrument was developed by researchers at the University of Texas Institute for Geophysics and fabricated by JPL, and it launched on October 14, 2024 atop a SpaceX Falcon Heavy rocket bound for Europa [3]. REASON operates at two frequency bands simultaneously, allowing it to distinguish between near-surface structure and deep basal reflections, a capability that no previous Europa-targeted instrument has demonstrated [1]. The instrument is flight-proven hardware with full mission-level heritage, placing it at TRL 9.

Europa Clipper carries REASON as part of a flyby science campaign: the spacecraft will loop past Europa roughly 50 times, collecting wide-swath regional maps of the ice shell from altitude [2]. That is exactly what Clipper is designed to do, and it will do it well. What Clipper is not doing is surface-level, high-resolution traverse profiling targeted at a specific candidate cryobot insertion site. There is no rover, no in-situ localization, and no onboard site-selection logic. The orbital geometry also limits any single flyby to a brief snapshot rather than a continuous along-track profile. My mission picks up where Clipper leaves off: a surface rover carrying an adapted REASON-heritage sounder drives a continuous traverse, building a spatially registered subsurface map at 100 m resolution or better, and autonomously flags the thinnest viable ice for a future cryobot deployment.

I. **What parameters are measured?**

REASON measures four primary parameterse. **Two-way travel time** (ns) is the elapsed time between the transmitted pulse and the received basal reflection. The raw measurement from which ice thickness is derived. **Signal amplitude at the basal reflector** (dB) captures the strength of the return from the ice-ocean interface. Returns too weak to distinguish from the noise floor are flagged and excluded from site-selection logic. **Bulk ice dielectric permittivity** (dimensionless) is estimated from the travel time and signal characteristics, where pure water ice sits around 3.15 and deviations indicate brine inclusions or heterogeneous layering. Finally, **derived ice shell thickness** (m) is computed onboard as $d = c \cdot t \,/\, 2\sqrt{\varepsilon_r}$, where $c$ is the speed of light, $t$ is the two-way travel time in seconds, and $\varepsilon_r$ is the estimated permittivity. This is the primary science output that feeds the subsurface map and the site-selection filter.

II. **What would a single data entry or record look like?**

The data product is organized as a multi-sheet spreadsheet with one row per radar trace, time-aligned across sheets by `trace_id`. Each record captures the full measurement context: radar sounder outputs, rover position, terrain conditions, and system health. A data quality flag (NOMINAL / CLUTTER / NOISE_FLOOR / INVALID) is attached to every trace so the onboard planner and downstream scientists can immediately distinguish usable science data from degraded returns.

*Example single record (Radar_Traces sheet):*

| Field | Value |
|---|---|
| trace_id | TR-0001 |
| sol | 1 |
| timestamp_s | 94.4 s |
| pos_x_m | 93.8 m (local ENU east) |
| pos_y_m | −7.1 m (local ENU north) |
| along_track_m | 94.0 m |
| twt_ns | 21647.3 ns |
| amplitude_dB | −41.62 dB |
| permittivity | 3.0624 |
| ice_thickness_m | 1855.5 m |
| tid_krad | 0.542 krad |
| quality_flag | NOMINAL |

III. **Who is the intended user of this data?**

The primary real-time consumer is the **onboard autonomous planning algorithm**. After each consensus round, the planner ingests the daily summary sheet to decide whether the traverse has produced a viable candidate site (ice thickness $<$5 km, slope $<20°$, roughness $<$0.7). If the criteria are met, it triggers a cryobot deployment recommendation; if not, it generates the next traverse waypoint. No human is in this loop. The secondary long-term consumer is **planetary scientists**: the full four-sheet data product, downlinked to Earth over the mission lifetime, provides a spatially registered along-track profile of ice shell thickness and permittivity variation across Europa's surface. This feeds directly into habitability assessments and informs the design of future subsurface access missions, extending the science value of the dataset well beyond the immediate site-selection objective.

**References:**

[1] NASA Jet Propulsion Laboratory, "Europa Clipper," JPL, Pasadena, CA. [Online]. Available: https://www.jpl.nasa.gov/missions/europa-clipper/. [Accessed: Jun. 16, 2026].

[2] Wikipedia contributors, "Europa Clipper," *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Europa_Clipper. [Accessed: Jun. 16, 2026].

[3] University of Texas at Austin Jackson School of Geosciences, "Scientific Instrument Developed by UTIG Scientists Headed to Europa," UT Austin, Austin, TX, Apr. 2025. [Online]. Available: https://www.jsg.utexas.edu/news/2025/04/on-oct-14-2024-nasas-europa-clipper-success [Accessed: Jun. 16, 2026].